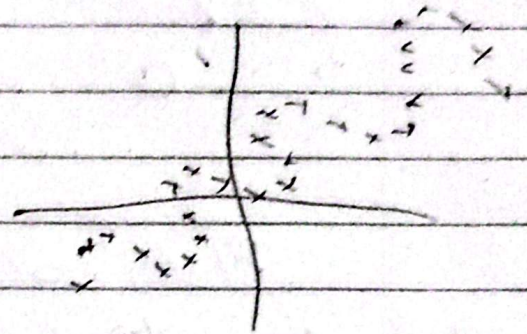


1) Gradient Boosting Regression.

Additive Modelling

$$y = x + \sin x$$



Training data: $\{(x_i, y_i)\}_{i=1}^n$

Differentiable loss function: $L(y, F(x))$

number of iterations: M .

~~Step 1:~~ $y = f(x)$

In ensemble:

$$f(x) = f_0(x) + \underbrace{f_1(x) + f_2(x) + \dots + f_M(x)}_{\text{Decision Tree}}$$

Step 1: Initialize $f_0(x) = \arg \min_Y \sum_{i=1}^N L(y_i, Y)$

$$\text{Let loss } f^0: L = \frac{1}{2} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Here, Y represent $\hat{y}_i$

$$\text{So, } f_0(x) = \arg \min_Y \frac{1}{2} \sum_{i=1}^N (y_i - Y)^2$$

$$\frac{\partial f(x)}{\partial r} = \frac{\partial}{\partial r} \left(\frac{1}{2} \sum_{i=1}^N (y_i - r)^2 \right)$$

$$\Rightarrow 0 = \sum_{i=1}^N (r - y_i)$$

Here solving for r gives the mean of output $[y_i]$

output of $f(x)$ will be mean.

Step 2: for $m = 1$ to $\underbrace{M}_{\text{no. of models}}$
 $f_1(x), \dots, f_M(x)$ if $M=50$

a) For $i = 1, 2, \dots, N$ compute.

$$r_{im} = - \left[\frac{\partial L(y_i, f(x_i))}{\partial (f(x_i))} \right]_{f=f_{m-1}}$$

$\underbrace{r_{im}}_{\text{pseudo residual error distance}} \quad \quad \quad \begin{aligned} &f_i := \text{pred output} \\ &f(x_i) = \text{predictive output} \end{aligned}$

$$r_{im} = - \left[\frac{\partial}{\partial y_i} \left(\frac{1}{2} (y - \hat{y}_i)^2 \right) \right]_{f=f_{m-1}}$$

$i = \text{row}$
 $m = \text{model no.}$

$f = f_{m-1}$
 previous answer.

$$r_{im} = (y_i - f(x_i))_{f=f_{m-1}}$$

For model 1 $[f_1(x)]$:

row 1: $r_{11} = y_1 - f_0(x_1)$

row 2: $r_{21} = y_2 - f_0(x_2)$

row 3: $r_{31} = y_3 - f_0(x_3)$

b) Fit a regression tree to the targets r_{jm} giving terminal regions R_{jm} , $j = 1, 2, \dots, J_m$.

~~to the other as input~~

Now consider output column as r_{11} .
 & decision tree regressor.

c) For $j = 1, 2, \dots, J_m$ compute

$$\hat{y}_j \leftarrow \underset{r}{\arg \min} \sum_{x \in R_{jm}} L(y_i, f_{m-1}(x_i) + r)$$

gradient boosting
value of leaf

Find the output value of the terminal value.
 gradient boosting

$$\frac{\partial L}{\partial r} = 0$$

We will get next output of the next model.